Five Conversations Shaping Nigeria's Upstream Digital Future

Operational Insights from SPE NAICE 2026 and What They Mean for Indigenous Operators

Omah Tech Enterprises Ltd.

August 2026

Executive Summary

Nigeria's upstream sector is entering a phase where operational discipline will determine which companies grow and which stagnate. As international oil companies continue divesting onshore and shallow-water assets, indigenous operators are inheriting older, more complex fields, less forgiving of inefficiency than the assets the industry's playbook was built on. Competitive advantage is no longer about portfolio size. It is about how well a company manages what it already has.

This paper draws on conversations held during SPE NAICE 2026 with professionals across the upstream value chain, including major operators, technology providers, and field-level engineers. Despite this range, the conversations converged on a consistent set of concerns. Production optimization remains difficult even with modern instrumentation. Mature assets generate more data than most organizations can meaningfully integrate into decisions. Core reservoir engineering workflows, including material balance and decline curve analysis, still consume days operational timelines increasingly cannot afford. Water production is emerging as a more urgent operational threat than generally acknowledged. And one theme recurred regardless of topic. Technology alone, without deep domain grounding, does not solve engineering problems.

Individually, these observations describe familiar operational friction. Together, they describe something more structural. Nigerian operators, particularly indigenous and marginal field operators with lean technical teams and tight margins, are not primarily short of data. They are short of integrated, timely decision support systems that turn existing data into engineering judgment they can act on. This gap is becoming a defining constraint on how quickly and profitably mature Nigerian fields can be operated. The regulatory environment reinforces this urgency. Nigeria's marginal field framework ties licence-holding to demonstrated production, and licences have already been revoked over prolonged non-production. For an indigenous operator, slow decision-making is no longer purely a cost problem. It can put the asset itself at risk.

For indigenous operators, this matters because inherited fields were not designed for their operating constraints, and the margin for slow decision-making has narrowed considerably. For regulators, it raises questions about how digital readiness should factor into asset transfer and development approval. For technology companies and investors, it points toward where value is likely to concentrate, in systems that accelerate engineering decisions grounded in the physical realities of Nigerian fields.

The five conversations behind this paper describe that shift, from asset acquisition toward operational and decision-making excellence, as something already visible in how operators are working today.

Introduction

Nigeria's upstream sector is undergoing one of its most significant ownership shifts in decades. International oil companies that once dominated onshore and shallow water production have spent the past several years divesting these assets, citing security risk, ageing infrastructure and a pivot toward deepwater operations. More than six billion dollars in onshore and shallow water assets have changed hands since 2021 alone, and indigenous operators, once minority participants, are now estimated to control more than half of the country's total oil production, according to PwC Nigeria's most recent sector outlook.

This transition is reshaping more than ownership registers. The fields changing hands are, in most cases, both technically mature and formally marginal, discovered decades ago and often produced inconsistently, now carrying the declining pressure and rising water cut of ageing reservoirs. Indigenous operators inheriting these assets face new pressures. Security costs, well abandonment liabilities and pipeline integrity issues once absorbed within a major's portfolio now sit directly on smaller balance sheets, while production must still be sustained with tighter access to capital and far less room for error.

The result is a sector where operational complexity is rising even as the pressure to control cost intensifies. Maximizing recovery from a mature well or deciding where to focus a limited drilling budget are no longer routine, but carry direct consequences for a company's financial position. Production optimization, once a quiet, background discipline, has become a central determinant of commercial performance. Digital technology sits at the center of this response because manual workflows and fragmented data are becoming less viable for managing increasingly complex assets. For operators running multiple ageing wells with lean teams, converting data into fast, reliable decisions is closer to a baseline requirement than a competitive advantage.

This was the backdrop for SPE NAICE 2026, held in Lagos in August 2026 under the theme "Thriving in the Evolving Global Energy Landscape: Collaborative Growth and Resilience," a theme reflecting an industry being asked to grow output, deepen local participation and withstand real pressure, all at once. Conversations across the conference, spanning major operators to field engineers, reflected this tension, and the underlying concerns were remarkably consistent regardless of who was speaking. Mature assets are producing complications, particularly around water management, that demand quicker diagnosis than most current workflows allow.

The five sections that follow examine these themes in detail, each building from a specific operational challenge into a broader industry observation grounded in evidence from across the sector. Together, they describe an industry in which the next competitive advantage will come down to the speed and quality of the engineering decisions operators make on the assets they already hold, more than the strength of new acreage.

Industry Observation 1

Production Optimization's Next Stage Is Real-Time

Production optimization is one of the oldest disciplines in upstream operations, and one of the most persistently difficult. It came up directly in a conversation with an engineer at Chevron's exhibition booth at SPE NAICE 2026. Asked which upstream problems he would most want technology to solve, he named inline solutions first, getting more value out of the constant stream of real time well data, and wellbore surveillance across a large well count second. He also mentioned Chevron's growing use of AI, including in data analysis, a sign that even an operator with mature digital infrastructure still sees more room to apply it. What stood out was that the underlying challenge, turning continuous data into a fast decision, remains real even for an operator with Chevron's resources.

Part of the difficulty is structural rather than technological. Optimizing a single well is straightforward. Optimizing a whole field, where wells share the same pipelines and processing equipment and each well's output affects the pressure the others operate under, is a much harder problem. Increasing output from one well today can accelerate water or gas breakthrough that reduces total recovery later, so what looks like optimization on a daily report can quietly work against the field's long term performance. That is why this kind of decision needs a single view spanning the reservoir, the well and the surface equipment together, built before the decision is made. Piecing three separate views together afterward, once something has already gone wrong, is a poor substitute.

The interest in inline solutions reflects a broader industry direction, whether or not every operator is there yet. Periodic well testing, still standard on many assets, only captures a snapshot of a well's performance every few weeks. A well that develops a temporary problem, a slug of water, a brief pressure drop, can behave normally during the scheduled test and still lose production in between without anyone noticing until the next check. Continuous monitoring closes that blind spot, and visibility is usually the first real constraint on fixing anything. The engineering judgment a well test provides still matters just as much once a problem is spotted, monitoring just gets it in front of someone faster.

Mature and marginal fields complicate this further. As a reservoir depletes and water cut rises, the right way to operate a well keeps shifting, a setting that worked well last year can quietly become the wrong one this year, and there is rarely a single obvious moment to change approach. Newly acquired marginal fields add a second complication. Historical records inherited from a previous operator are often incomplete, so the real work often starts with rebuilding a reliable production history before any forward looking decision can be made with confidence.

For a major with Chevron's scale, the basic problem of pulling data together has likely already been solved. That may be why the conversation moved quickly toward AI instead of integration. For most indigenous and marginal field operators, that same problem is nowhere close to solved. Pressure data, well test results, surveillance logs and production history typically live in separate systems on separate schedules, when they are tracked consistently at all. When those systems do not talk to each other, someone ends up reconciling them by hand, since there is no other way to see a well's current status in one place.

This is where digital technology has the clearest role to play, closing that lag without taking engineering judgment out of the loop. Automated surveillance that flags an unusual pressure trend or a shift in water cut the day it happens, instead of whenever an engineer next has time to check each well by hand, changes what is operationally possible. An engineer still has to interpret the signal and decide what to do about it. What actually changes is the delay between the signal appearing and someone qualified seeing it, which for a field running on thin margins is often the difference that matters most.

Key Insight

Production optimization has always been difficult, but the nature of the difficulty is changing. The old constraint was measurement, not knowing enough about a well to act on it. Increasingly, the constraint is workflow, having the data but no fast enough way to turn it into a decision. That shift matters most for operators running many mature wells without the systems a major already has in place. For a field running on thin margins, closing that gap can be the difference between catching a problem early and losing weeks of production to one that goes unnoticed.

Industry Observation 2

Mature Assets Need Integration, Not More Tools

A conversation with a reservoir engineering academic and researcher at SPE NAICE 2026 approached the same problem from a different angle. Asked where the best technology opportunities in Nigerian reservoir engineering lay, his answer came explicitly from a research and technology development perspective rather than a field operations one, and he was clear about that distinction, encouraging continued direct engagement with field operators to surface their operational pain points as a separate, necessary track. His own answer kept returning to one idea. Bringing production data, pressure readings, well tests, surveillance logs and reservoir simulation together into one system built to support real time decisions creates far more value than digitizing any of those individually. For a marginal field operator, that kind of integration is what separates a fast, confident decision about re-entering an old well or where to place a new one from a slow, uncertain one.

Even where each individual data stream is available quickly, the harder problem is what happens when they are looked at separately. A pressure reading might show a decline that looks like normal reservoir depletion. A production log from the same period might show a steeper drop than that decline alone would explain. Neither one, on its own, tells an engineer whether the well has a mechanical problem or is simply behaving as expected. The answer only becomes clear once pressure, production, well test results and the simulation model are checked against each other side by side, and that is exactly the step isolated tools do not support and manual cross checking makes slow.

Integration has a second dimension beyond data. The same conversation pointed to a related idea, optimizing the reservoir, the well and the surface equipment together instead of one after another. A reservoir model might flag a well as a strong candidate for higher output. Whether that actually makes sense depends on whether the well itself can handle the extra flow without pulling in more water, and whether the surface equipment downstream has room to take it without slowing the rest of the field down. Answering those three questions separately, often by different specialists using different tools, is how a technically sound recommendation quietly becomes unworkable by the time it reaches the field.

AI assisted surveillance fits directly into this need, doing that same cross checking automatically and continuously instead of by hand and only occasionally. A system already holding pressure, production, well test and simulation data in one place is well placed to notice when those readings stop agreeing with each other, often the earliest sign of a developing problem. Water production is a good example. A rising water cut can come from a few different causes, and telling them apart depends on this same kind of cross checking rather than watching the water cut number alone, a distinction this paper comes back to later.

None of this is well served by isolated software tools, however good each one is on its own. A strong decline curve package, a strong well test package and a strong simulator can each be excellent individually and still leave an engineer doing the actual integration by hand, checking one file against another and reconciling the two in a spreadsheet. For a well resourced asset with a large technical team, that manual step is an inconvenience. For a marginal field operator managing several ageing wells with a handful of engineers, it is often the difference between catching a developing problem this month and finding it three months later.

That timing gap matters more for marginal field operators than it would elsewhere. Nigeria's marginal field framework ties continued licence holding to demonstrated production, which leaves less room than a better resourced operator might have for a developing problem to sit unaddressed.

Key Insight

Mature Nigerian fields need a connective layer between the tools and data sources that already exist, pressure, production, well tests and simulation, brought into one place where they can be checked against each other quickly, more than they need another standalone analytical tool. For a marginal field operator, that matters beyond convenience. Nigeria's marginal field framework ties continued licence holding to demonstrated production, which leaves less room than usual for a developing problem to go unaddressed.

Industry Observation 3

Reservoir Engineering Needs Hours, Not Days

The same conversation with a reservoir engineering academic and researcher at SPE NAICE 2026 that raised the case for integrated decision support also touched on a narrower, more specific frustration. Even where an engineer has all the data required for a piece of core reservoir engineering analysis, material balance, decline curve analysis, pressure test interpretation, PVT analysis, building an optimization model, the analysis itself still typically takes days rather than hours. These are the routine backbone of reservoir engineering work, and the time they consume is a real, recurring cost.

That time rarely goes to the part requiring the most expertise. A material balance study, for instance, needs a clean, reconciled production and pressure history before the calculation itself can even begin, and assembling that history is usually the slowest part of the exercise, not the calculation itself. The same pattern holds across decline curve analysis, pressure test interpretation, PVT analysis and optimization modelling. Each depends on properly prepared data before the technique can be applied with any confidence. The genuinely hard part, the judgment call about which model fits and what a result actually means, usually takes a fraction of the total time. Most of the days involved go to preparation and cross checking, necessary work that does not require a senior engineer's training to perform.

AI has a role here, but a narrow one. It does not replace the engineer's interpretation. A well test can often be matched by more than one plausible model, and choosing between them depends on context an algorithm does not have, how nearby wells have behaved, what the geology suggests, how much risk the operator is willing to carry on a given decision. What AI can reasonably do is compress the preparation stage, cleaning and structuring data, flagging likely outliers, producing a first pass fit for a qualified engineer to review and approve rather than build from a blank sheet. The engineer still makes the call. They simply spend their time making it rather than preparing to make it.

For a well staffed asset, this mostly changes how a day is spent. For a marginal field operator with a handful of engineers covering many wells, it changes what is possible at all. An engineer who can complete a material balance review in an afternoon instead of a week can reasonably cover several wells across a month. One who cannot is, in practice, choosing which wells get analyzed and which do not, since the technical work required to look properly is not something a small team can do for every well, every time.

Key Insight

The slowest parts of core reservoir engineering work are rarely the parts that need an engineer's judgment most. Assembling and cleaning data, fitting a first pass model, checking for obvious inconsistencies, these consume the bulk of the time and do not require years of training to do well. What does require that training is deciding what a result actually means once it is in front of you. The clearest role for AI in this work is narrowing the gap between data and a first, defensible analysis, not narrowing the role of the engineer who has to stand behind the conclusion.

Industry Observation 4

The Real Cost of Water Breakthrough Is Time

Water production came up in nearly every field level conversation at SPE NAICE 2026, but one description was specific enough to be worth building a section around. An operator described a well producing almost entirely water from one production string, with only a limited amount of oil still coming through. The well was not unproductive in the sense of being shut in or abandoned. It was producing steadily. Most of what it was producing was simply not oil anymore.

This is not a rare situation. Water cut rises on nearly every well eventually, as reservoir pressure depletes and the balance between oil and water in the produced fluid shifts. What makes it a sharper problem on Nigeria's mature and marginal fields specifically is timing. Many of these fields, as the introduction to this paper noted, were discovered decades ago and often produced only inconsistently before being reallocated, which means a number of them are inheriting an advanced stage of water production almost from the point of reacquisition, before an operator has had the chance to build the technical and financial cushion a longer, more gradual decline would normally allow.

A rising water cut, on its own, does not say what is causing it, and the appropriate response differs completely depending on the answer. Water coning, where excessive drawdown pulls the underlying water table up toward the perforations, often responds to a straightforward operational change, reducing drawdown or adjusting the choke, sometimes solving most of the problem within days. Water channeling behind casing, caused by a cement or completion integrity issue, does not respond to that kind of adjustment at all. It typically requires a workover, a materially larger cost, and simply reducing drawdown may do little except reduce total production without addressing the water at all. A rising contribution from an advancing aquifer is different again, and usually calls for a longer term production strategy rather than a single fix. Three different root causes, one shared symptom, and three very different bills depending on which one an operator is actually looking at.

Distinguishing between them depends on the same kind of integrated view described earlier in this paper, production history, pressure trends and well test results checked against each other rather than watched in isolation. A water cut chart on its own usually cannot make that distinction. The pattern of how water cut has moved against pressure and production over time usually can.

Timing is where the real cost sits. A water cut trend caught early, while it is still a gradual shift, usually leaves the full range of responses available, including the cheapest ones. The same trend caught late, after oil production has already fallen sharply, tends to have foreclosed the cheaper options by the time anyone looks closely, leaving a workover or a more expensive intervention as the only paths left. The technical diagnosis does not become harder with time. The range of affordable responses to it gets narrower.

For a marginal field operator, that narrowing has consequences beyond the immediate cost of the fix. A well quietly producing mostly water is a well quietly not meeting the production expectations tied to that operator's licence, a link this paper's introduction already noted. Diagnosing the problem quickly enough to still have a cheap option available is not simply a matter of protecting a single well's output. On the margins these operators work with, it can be part of protecting the asset itself.

Key Insight

Water breakthrough is treated as a production problem, but the costliest part of it is usually a diagnosis problem. The same rising water cut can call for a cheap adjustment or an expensive workover, and telling those apart depends on catching the trend early enough for the cheaper option to still be on the table. Once the diagnosis is available, the fix itself is rarely the hard part.

Industry Observation 5

Technology Alone Does Not Solve Engineering Problems

A conversation with an engineer who spent years at an indigenous exploration and production company returned repeatedly to one point. Domain credibility is what actually separates technology built for this industry from technology that only resembles it. His own path, from field level petroleum engineering into building a company of his own, is itself an example of the pairing this section is about. He has stood on both sides of the problem, the engineering and the building.

Building software that displays production numbers clearly is, in relative terms, the easier problem. It requires competent software development and nothing more, no understanding of what a declining well actually looks like or what a rising water cut might mean. Building something that genuinely helps an engineer make a faster, better decision is a different task entirely. It requires understanding the reservoir engineering underneath the numbers well enough to know which patterns matter and which are noise. A team with only the first kind of skill can ship a technically functional product. Whether that product is actually useful to the engineer relying on it is a separate question, one that only the second kind of skill can answer.

Neither skill set is sufficient alone. A petroleum engineer without software development ability can describe exactly what a good decision support system should do, and has no way of building it at any scale. A software engineer without reservoir engineering understanding can build something technically polished that quietly gets the physics wrong, flagging a pressure trend as concerning when it is actually normal depletion, or missing a genuine early warning because it does not match a pattern the system was built to recognize. Every example already covered in this paper needed both. Knowing which inline signal actually mattered took an understanding of the reservoir behind the signal. Compressing reservoir engineering prep time without losing the judgment call at the end of it took the same pairing. So did building an integration layer that knows what to check against what because it understands the physics, not just the data format, and telling water coning apart from channeling behind casing, which needs software able to hold that physical distinction rather than just show numbers on a screen. None of it works with only one half of the pairing in the room.

Credibility in this space is built through direct engagement with operators, not through general technical capability alone. A system built without real exposure to how field engineers work, what data they trust, and what they routinely have to work around tends to solve a cleaner, more theoretical version of the problem than the one operators actually face. The domain knowledge that matters here is not solely academic reservoir engineering training. It includes an understanding of field conditions, workflow constraints and operator priorities that comes only from working closely with the people doing the job.

Key Insight

Software that displays data cleanly is not hard to build. Software that helps an engineer make a better decision requires understanding the reservoir engineering underneath the numbers, built by people who have spent real time with the operators doing the work. That combination is what makes a technology company credible in this industry.

Emerging Industry Pattern and Implications for Nigeria

Emerging Industry Pattern

Five different conversations, five different starting points, and the same underlying issue kept surfacing. In each case, the technical data needed to make a good decision already existed somewhere. What was missing was a fast, trustworthy way to turn that data into a decision an engineer could act on. Production optimization's constraint has shifted from measurement to workflow. Mature assets are missing the connective layer that ties their existing tools together. Core reservoir engineering techniques spend most of their time on data preparation, with the actual judgment call taking only a fraction of it. Water breakthrough gets expensive mainly through late diagnosis. And credible technology in this sector depends on domain understanding as much as software skill, because a tool built without that understanding tends to solve a cleaner version of the problem than the one an engineer actually faces.

None of these five stories was really about a shortage of data. Each one turned on whether engineering judgment, applied quickly enough to matter by people who understand the reservoir behind the numbers, was actually available when it counted. For Nigeria's mature and marginal fields specifically, where licence terms are tied to demonstrated production and the margin for a slow decision is thin, the operators who close that gap fastest are the ones best positioned to keep the asset, not just improve the margin.

Implications for Nigeria

Indigenous operators carry the most immediate stake in this, particularly where a licence depends on demonstrated production. Closing the decision support gap is closer to a core operating requirement than something to get to eventually. Regulators have a real question worth asking too, even if it has not typically been part of the conversation. Should digital readiness sit alongside reserves and capital commitments in how asset transfers and development timelines get evaluated?

Observation Five offers the clearest guide for technology companies and the developers building for this sector. Teams without real domain credibility tend to build tools that work cleanly in a demo and struggle to earn trust once they reach the field. Investors weighing this space should read the same signal. Teams pairing engineering depth with software capability are likely to hold up better over time than teams with only one or the other.

Universities and the engineering graduates they train have the longest runway to respond, but the direction is already visible in everything above. Data and software literacy are becoming part of what reservoir and production engineering training itself needs to cover.

Future Outlook

If the patterns described in this paper hold, three developments are likely to define the next phase of digital transformation in Nigeria's upstream sector.

The first is the digital twin, a live, continuously updated model of a field's reservoir, wells and surface network together, rather than the static, periodically rebuilt models most operators still work with. Much of what this paper has described, integrating disparate data sources, checking pressure against production against simulation, coordinating reservoir and well and surface decisions, are the working components of a digital twin, encountered here as separate problems rather than one connected capability. For a marginal field operator, a full digital twin in the sense large operators use the term may be some way off. The direction it points toward, a single continuously current picture of the asset rather than several partial ones, is available much sooner and is where the near term value sits.

The second is AI assisted reservoir management extending beyond individual analyses into something closer to continuous surveillance, models that update as new data arrives rather than being rebuilt each time an engineer has reason to look. Combined with the faster reservoir engineering workflows described earlier in this paper, a well's status becomes something an engineer checks and confirms rather than something reconstructed from scratch each time a question arises.

The third is a genuine shift in what production surveillance and field optimization mean day to day. At present, these remain something operators do periodically, on a schedule dictated by available time and staff rather than by what the reservoir needs. As the tools described throughout this paper mature, surveillance and optimization become closer to continuous, less a task scheduled around everything else and more a standing part of how a field is run.

None of this argues for more dashboards. A dashboard displays what is already known, and nothing in this paper suggests Nigerian operators are short of things to look at. What the industry has consistently signalled, across five separate conversations, is a need for systems that help engineers decide what to do next, and do it fast enough to matter. The technology companies that build for that need are the ones likely to matter most in Nigeria's upstream sector over the next several years.

Conclusion

The evidence gathered across five conversations at SPE NAICE 2026 points toward a single conclusion. Nigeria's upstream sector is entering a period where the companies that grow fastest will not necessarily be the ones that acquire the most acreage. They will be the ones that make better operational decisions on the assets they already hold, faster, and with a clearer understanding of the reservoir behind every number.

This is not a call to abandon growth through acquisition. Indigenous operators will keep acquiring assets as majors continue to divest, and that trend shows no sign of slowing. What this paper has argued is narrower, and in some ways more urgent. Acquiring an asset is only the first half of the work. The fields now moving into indigenous hands are mature, often marginal, and increasingly tied to production commitments that leave little room for slow decisions. Whether an operator can act quickly and correctly on what its data is telling it may end up mattering as much as which assets it holds in the first place.

The conversations behind this paper, spanning a major operator, a reservoir engineering researcher, and engineers working directly on marginal fields, suggest this shift is not theoretical. It is already shaping how the industry thinks about production optimization, reservoir engineering workflows, water management and what makes a technology company credible in this space.

These observations sit at the center of what OmahTech does, a technology company that bridges the gap between traditional petroleum engineering and applied machine learning to build data-driven decision support systems for upstream oil and gas operations helping Nigeria's indigenous operators get more out of the assets they already hold. This paper is offered in that spirit, as one contribution to a conversation the industry will need to keep having as more of these fields change hands.

References

1. Federal Republic of Nigeria. Petroleum Industry Act, 2021.
2. "NUPRC and OML 46 Revocation." ThisDay, October 11, 2022. https://www.thisdaylive.com/index.php/2022/10/11/nuprc-and-oml-46-revocation/
3. PwC Nigeria. "Oil & Gas Sector Performance and Outlook." Presented at the Lagos Chamber of Commerce and Industry Economic Outlook, January 2026. https://www.lagoschamber.com/wp-content/uploads/2026/01/Oil-Gas-Sector-Performance-and-Outlook-FV.pdf
4. Society of Petroleum Engineers Nigeria Council. SPE NAICE 2026, Nigeria Annual International Conference and Exhibition. Eko Hotels and Suites, Victoria Island, Lagos, August 3 to 5, 2026. https://home.spenigeriacouncil.org/